Adding Polygons

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2. Implementation Effect
3. Implementation Code

1. Implementation Principle

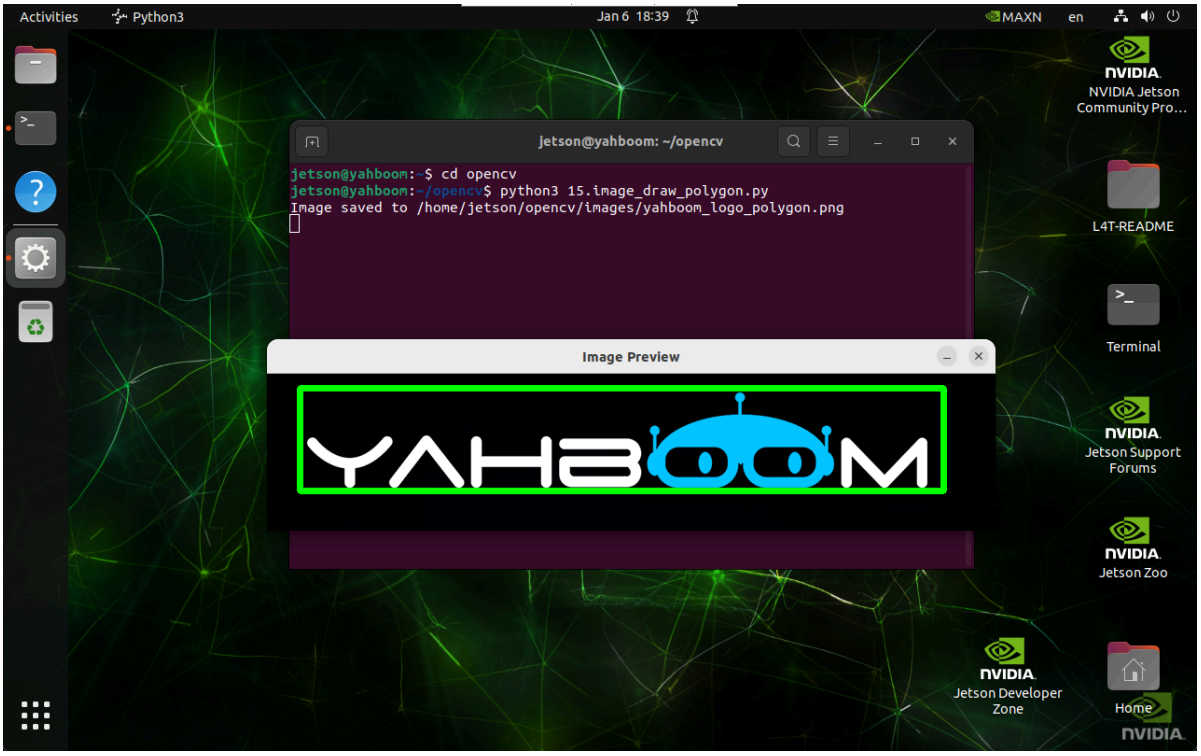
Use the `cv2.polylines()` function to draw a polygon on the image.

2. Implementation Effect

```
cd ~/opencv
```

```
python3 15.image_draw_polygon.py
```

> Note: click the image window and press any key to exit.



3. Implementation Code

```
import cv2
import numpy as np

def draw_polygon(input_path, output_path, points, is_closed, color, thickness):
    image = cv2.imread(input_path)
    if image is None:
        print("Error: Unable to open image file.")
        return
    points = np.array(points, np.int32)
```

```
    points = points.reshape((-1, 1, 2))
    cv2.polylines(image, [points], is_closed, color, thickness)
    if cv2.imwrite(output_path, image):
        print(f"Image saved to {output_path}")
        cv2.imshow('Image Preview', cv2.imread(output_path))
        cv2.waitKey(0)
        cv2.destroyAllWindows()
    else:
        print("Error: Unable to save image file.")

draw_polygon('/home/jetson/opencv/images/yahboom_logo.png', \
           '/home/jetson/opencv/images/yahboom_logo_polygon.png', \
             [(35, 15), (725, 15), (725, 125), (35, 125)], True, (0, 255, 0), 5)
```